

PROBLEM B: The Airplane Seating Problem

Airlines are free to seat passengers waiting to board an aircraft in any order whatsoever. It has become customary to seat passengers with special needs first, followed by first-class passengers (who sit at the front of the plane). Then coach and business-class passengers are seated by groups of rows, beginning with the row at the back of the plane and proceeding forward.

Apart from consideration of the passengers' wait time, from the airline's point of view, time is money, and boarding time is best minimized. The plane makes money for the airline only when it is in motion, and long boarding times limit the number of trips that a plane can make in a day. The development of larger planes, such as the Airbus A380 (800 passengers), accentuate the problem of minimizing boarding (and deboarding) time.

Devise and compare procedures for boarding and deboarding planes with varying numbers of passengers: small (85–210), midsize (210–330), and large (450–800).

Prepare an executive summary, not to exceed two single-spaced pages, in which you set out your conclusions to an audience of airline executives, gate agents, and flight crews

Note: The 2 page executive summary is to be included in addition to the traditional model description.

An article appeared in the NY Times Nov 14, 2006 addressing procedures currently being followed and the importance to the airline of finding better solutions. The article can be seen at <http://travel2.nytimes.com/2006/11/14/business/14boarding.html>.

B 题：飞机就座问题

航空公司允许引领候机乘客以任何次序就座。已经成为惯例的是首先引领有特殊需要的乘客就座，然后是头等舱的乘客就座(他们坐在飞机的前部)。然后引领持经济舱和商务舱机票的乘客从飞机后排开始向前按照排结组就座。

从航空公司的角度来看，除了考虑乘客的等候时间外，时间就是金钱，所以登机时间最好要减到最少。飞机只有在飞行的时候才能为航空公司赚钱，而长的登机时间限制了一架飞机一天中可以飞行的次数。

诸如 Airbus A380(空中客车 A380，可以容纳 800 名乘客)的大型机的发展就更要强调缩短登机(以及下机)时间的问题了。

就乘客人数不同的飞机：小型机(85-21)，中型机(210-330)和大型机(450-800)，设计登机和下机时间的步骤并进行比较。

准备一份不超过两页纸(不空行打印)的实施概要，以便向航空公司业务主管、登机口执法人员以及空(地)勤人员阐明你们的结论。

在 2006 年 11 月 14 日的《纽约时报》上刊登的一篇文章报告了当前遵循的步骤以及航空公司寻求更好的解决方案的重要性。该文可以在如下网址找到：
<http://travel2.nytimes.com/2006/11/14/business/14boarding.html>